

**Phishing Attack and an Employee Account Compromise Investigation at
Dansoman Community Bank**

Capstone Final Report — Incident Investigation Track

Organization: Dansoman Community Bank (fictional)

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1. Executive Summary

This project investigated a phishing attack against a fictional community bank, Dansoman Community Bank, in which a Finance Department employee's account and computer were compromised after the employee clicked a link in a fake IT-support email. The compromised account was then used to log into the bank's core banking system in about twelve minutes. Using email, login, and network records, the investigation traced exactly how the attack unfolded and found that the bank had no written security policy in place at the time to guide staff behavior or the response. The most important finding is that a single successful phishing email was enough to reach the bank's most sensitive system, because there was no second layer of protection, such as multi-factor authentication, requiring more than a password to log in. As a result of this investigation, the bank adopted a set of written security policies and technical controls to close these gaps.

2. Background and Scope

Dansoman Community Bank is a fictional community bank offering savings and business banking, money transfers, and digital financial services, used as the basis for this project. The scenario examined is a single phishing incident affecting one Finance Department employee, jmensah, reconstructed from email, authentication, endpoint, and network logs covering the period the attack occurred. This project is scoped to analyzing this specific incident — its timeline, impact, and response — and does not include a full assessment of the bank's wider security infrastructure beyond what is directly relevant to this incident.

3. Incident Investigation

Incident Timeline

The timeline below was reconstructed by correlating email, authentication, endpoint process, and network connection logs for the affected employee and workstation. At least one stage is mapped to a MITRE ATT&CK technique, as shown in the third column.

Time	Event	ATT&CK Reference
09:12	jmensah receives an email spoofed to appear from IT Support, requesting an urgent browser security update	T1566.002—Phishing: Spearphishing Link (Initial Access)
09:13	jmensah clicks the link contained in the email	T1204.001—User Execution: Malicious Link (Execution)
09:14	A failed login attempt for jmensah's account is recorded from external IP 41.2.3.10	T1078 — Valid Accounts (Initial Access / Defense Evasion)
09:14:47	A successful login for jmensah's account is recorded from the same external IP, indicates credential compromise	T1078 — Valid Accounts
09:15–09:16	The workstation establishes two outbound connections to unfamiliar external address 185.9.2.1	T1071 — Application Layer Protocol (Command and Control)

Time	Event	ATT&CK Reference
09:17	A suspicious executable, BrowserUpdate.exe, appears on the workstation and is executed	T1204.002 — User Execution: Malicious File (Execution)
09:18	A scheduled task, WinUpdateHelper, is created and started, indicating an attempt to maintain access	T1053.005 — Scheduled Task (Persistence)
09:22–09:24	jmensah's account accesses the Finance and HR servers, then attempts to access the Core Banking Server	T1078 — Valid Accounts (Lateral Movement)
09:24:51	A successful login to the Core Banking Server is recorded under jmensah's account	T1078 — Valid Accounts
09:25	The workstation communicates with 185.9.2.1 again	T1071 — Application Layer Protocol
09:26	The WinUpdateHelper scheduled task remains active	T1053.005 — Scheduled Task

Root Cause

The root cause was the absence of any written security policy or technical control requiring more than a password to reach sensitive systems. jmensah's password alone was sufficient both to compromise the account at 09:14:47 and to log into the Core Banking Server at 09:24:51 — no additional authentication factor was required at either point. This was compounded by the absence of email filtering that could have flagged the spoofed IT-support sender before the message reached jmensah's inbox at 09:12, and by the absence of endpoint controls that could have blocked BrowserUpdate.exe from executing at 09:17. Because no documented incident response procedure existed, there was also no defined trigger for escalating the failed-then-successful login at 09:14 before the attacker reached the Core Banking Server ten minutes later.

Impact

- Confidentiality: jmensah's account credentials were exposed and used by an unauthorized party; the Finance and HR servers were accessed between 09:22 and 09:24, and the Core Banking Server was accessed at 09:24:51, placing customer account, transaction, and employee data at risk of unauthorized viewing.
- Integrity: the successful login to the Core Banking Server created a risk of unauthorized changes to customer records or transactions; a post-incident review of activity during the compromise window found no confirmed unauthorized transactions.
- Availability: no disruption to system availability was recorded in the logs during the incident; the affected workstation was later taken offline as part of remediation.

Response Actions Taken

- jmensah's account was disabled to stop further use of the compromised credentials.
- The affected workstation was isolated from the network, and external IP addresses 41.2.3.10 and 185.9.2.1 were blocked at the perimeter.

- BrowserUpdate.exe and the WinUpdateHelper scheduled task were removed, and the workstation was confirmed clean before being restored from a known-good image.
- jmensah's credentials were reset and the account was enrolled in multi-factor authentication before being re-enabled.
- All activity performed on the Finance, HR, and Core Banking servers during the compromise window was reviewed for unauthorized changes or transactions.
- The account was monitored closely for 14 days after restoration.

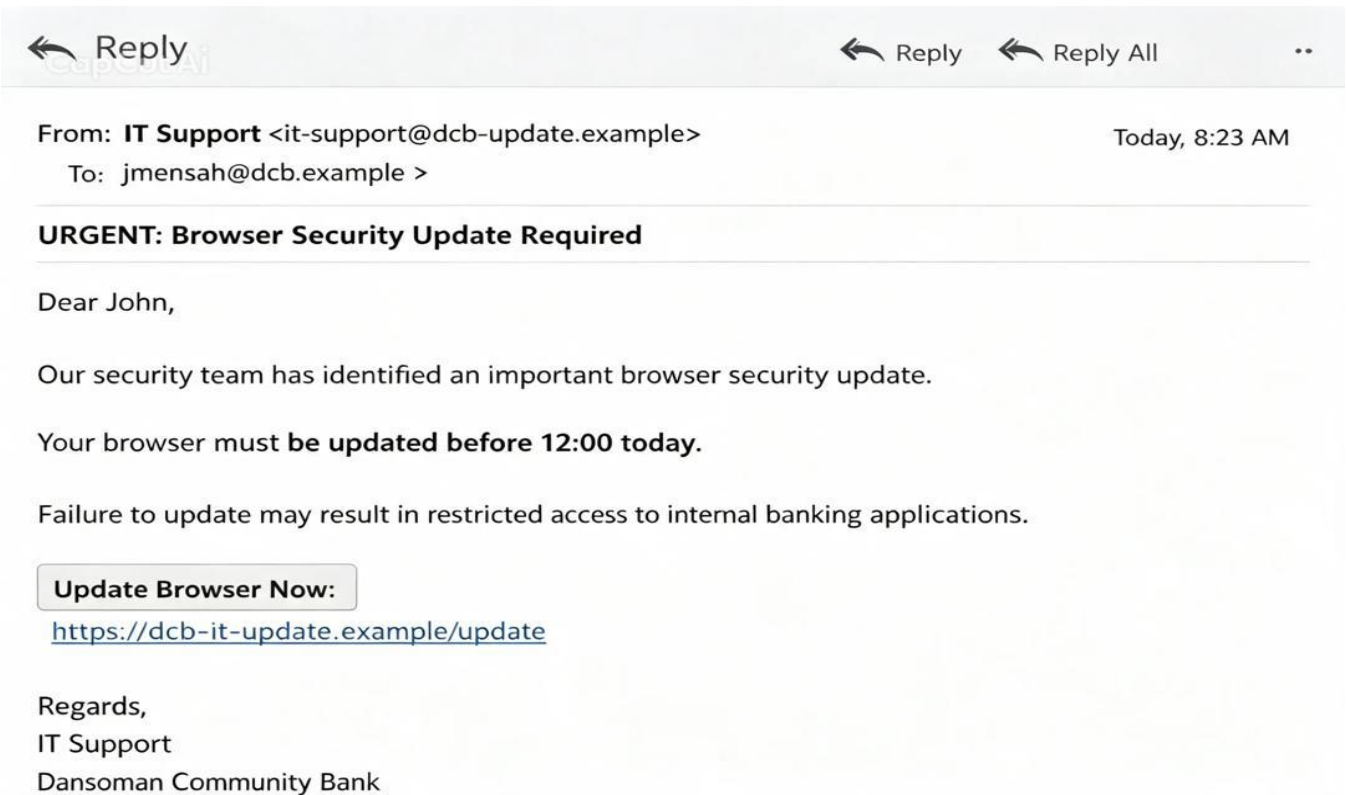
4. AI Integration

Co-Pilot was used to generate the raw log sources and the phishing email screenshot. Claude (Anthropic's AI assistant) was used as an AI-assisted analysis tool to correlate the separate raw log sources for this incident into a single, ordered timeline. Each log source used a different timestamp format and set of fields; the AI tool was used to cross-reference timestamps, the employee's account identifier, and the relevant IP addresses across all sources so that the events could be placed in a single correct sequence, and to flag the points in that sequence where behavior changed meaningfully.

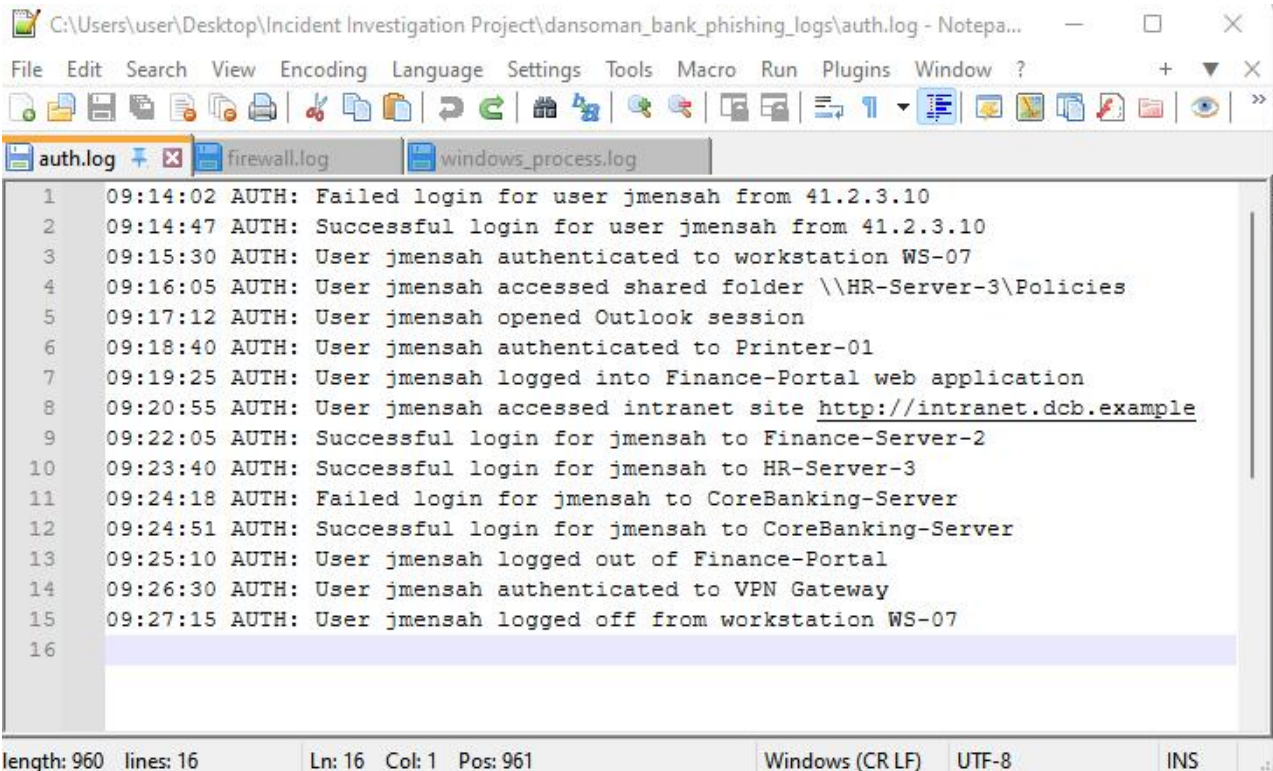
5. Recommendations

- Adopt a formal, documented security policy suite for the bank, including an Incident Response Policy, Acceptable Use Policy, Access Control Policy, and Data Classification Policy, so staff and responders have defined procedures to follow.
- Require multi-factor authentication on all remote logins and specifically on the Core Banking Server, so a compromised password alone cannot grant access.
- Deploy email filtering with sender authentication checks and link inspection to catch spoofed "IT support" style emails before they reach staff inboxes.
- Deploy endpoint application controls (allow-listing or similar) to block unsigned or unrecognized executables such as BrowserUpdate.exe from running.
- Add real-time alerting for logins from unfamiliar external IP addresses and for newly created scheduled tasks, to shorten the roughly ten-minute detection gap observed in this incident.
- Extend Wazuh integration to cover log sources that were not monitored during this incident, particularly the email gateway (the spoofed IT-support email at 09:12 generated no alert) and the Core Banking Server and network firewall directly, rather than relying solely on endpoint and authentication logs, so that future attacks are visible at every stage, not only after a workstation is already compromised.
- Segment the network so that user workstations are isolated from the Finance, HR, and Core Banking servers. As shown in the network diagram in Section 6, the laptop and the Ubuntu Server hosting all three systems currently sit on the same 10.10.10.0/26 segment behind a single switch; placing the servers on a separate VLAN with firewall rules restricting which hosts can reach them would have prevented jmensah's compromised workstation from reaching the Core Banking Server directly.
- Run refresher security awareness training and simulated phishing exercises for staff, focused on recognizing spoofed IT-support communications.

6. Evidence



Phishing email screenshot



Authentication Logs

C:\Users\user\Desktop\Incident Investigation Project\dansoman_bank_phishing_logs\firewall.log - Notep...

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auth.log firewall.log windows_process.log

```
1 09:15:10 FIREWALL: ALLOW SRC: 10.10.10.129 DST: 185.9.2.1 PORT: 4444 PROTO: TCP
2 09:16:32 FIREWALL: ALLOW SRC: 10.10.10.129 DST: 185.9.2.1 PORT: 4444 PROTO: TCP
3 09:17:05 FIREWALL: BLOCK SRC: 10.10.10.129 DST: 198.51.100.23 PORT: 23 PROTO: TCP
4 09:18:47 FIREWALL: ALLOW SRC: 10.10.10.129 DST: 185.9.2.1 PORT: 4444 PROTO: TCP
5 09:20:12 FIREWALL: BLOCK SRC: 10.10.10.129 DST: 203.0.113.45 PORT: 22 PROTO: TCP
6 09:22:59 FIREWALL: ALLOW SRC: 10.10.10.129 DST: 185.9.2.1 PORT: 4444 PROTO: TCP
7 09:25:30 FIREWALL: ALLOW SRC: 10.10.10.129 DST: 185.9.2.1 PORT: 4444 PROTO: TCP
8 09:26:14 FIREWALL: BLOCK SRC: 10.10.10.129 DST: 192.0.2.77 PORT: 3389 PROTO: TCP
9
```

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Firewall Logs

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auth.log firewall.log windows_process.log

```
1 09:17:15 WINDOWS: File created: C:\Users\jmensah\Downloads\BrowserUpdate.exe
2 09:17:19 WINDOWS: BrowserUpdate.exe executed.
3 09:17:21 WINDOWS: Process initiated outbound network connection.
4 09:18:20 WINDOWS: Scheduled task created: WinUpdateHelper
5 09:18:23 WINDOWS: WinUpdateHelper started.
6 09:19:05 WINDOWS: User jmensah opened Microsoft Word document: C:\Users\jmensah\Do
7 09:19:47 WINDOWS: Print job submitted by jmensah to Printer-01
8 09:20:12 WINDOWS: Outlook.exe launched by user jmensah
9 09:21:30 WINDOWS: Email sent from jmensah@dcb.example to hr@dcb.example
10 09:22:10 WINDOWS: System idle state entered
11 09:24:02 WINDOWS: User jmensah accessed shared folder \\Finance-Server-2\Reports
12 09:25:44 WINDOWS: Excel.exe opened file: C:\Users\jmensah\Desktop\Budget2026.xlsx
13 09:26:00 WINDOWS: WinUpdateHelper remains active.
14 09:27:15 WINDOWS: User jmensah logged off from workstation
15
```

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Sysyem Logs

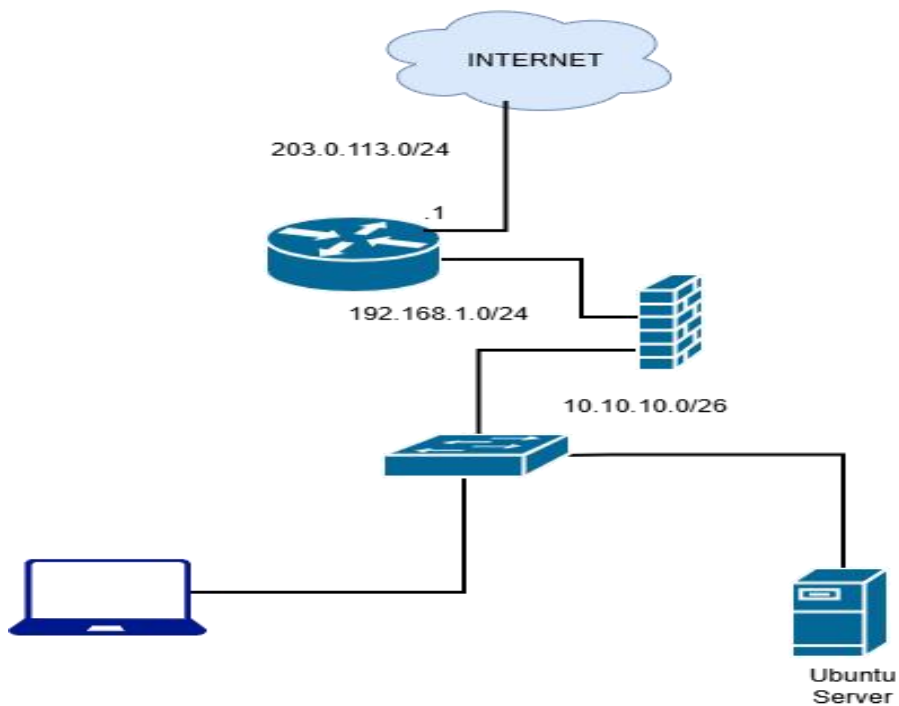


Diagram: Network architecture showing the path from jmensah's workstation to the Finance, HR, and Core Banking servers before segmentation.

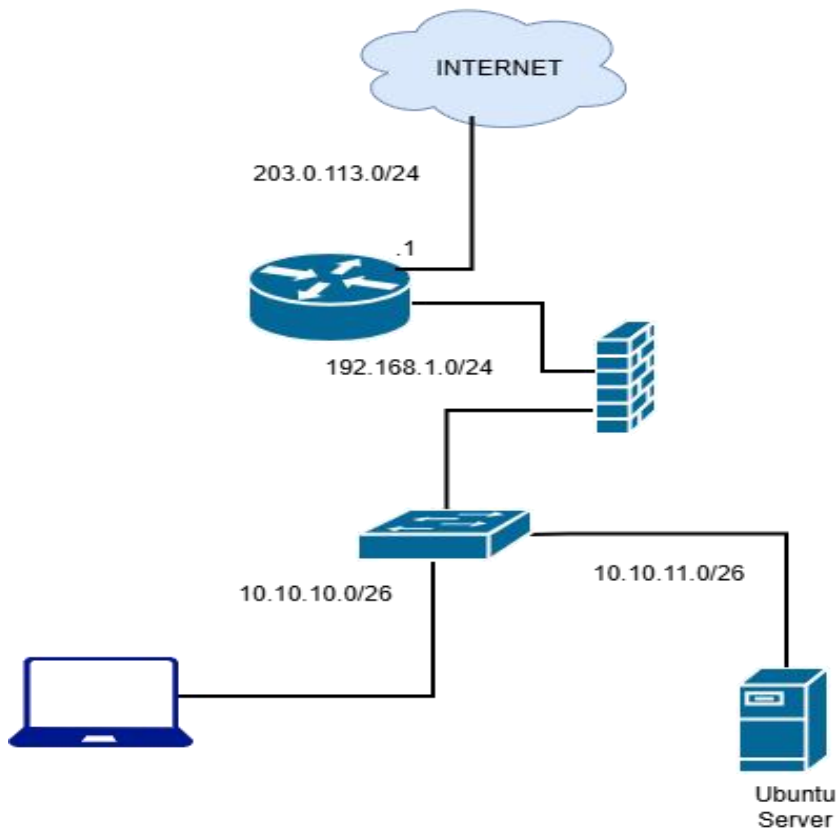


Diagram: Network architecture after segmentation.

7. Reflection

This project relied most on the log correlation and timeline-reconstruction skills from Week 3, and the incident response framework from Week 5, which together made it possible to turn raw log sources into a single clear account of the attack and a structured response. If starting again, I would build in a way to check for signs of data exfiltration or further lateral movement beyond the Core Banking Server login itself, since the available logs ended while the attacker's persistence mechanism was still active.